# Where Does the Name AdaBoost Come From?

AdaBoost is short for **Adaptive Boosting**. It was introduced by Yoav Freund and Robert Schapire in 1995 [6] (published in 1997 [7]), work that later earned them the 2003 Gödel Prize [8].

"Boosting" was already the name for a broader idea. In the late 1980s, Michael Kearns and Leslie Valiant asked whether a "weak learner," something only slightly better than random guessing, could be combined with others into a "strong learner" that is highly accurate [1, 2, 3]. This question was posed within the PAC learning framework [9]. Schapire showed in 1990 that it could [4], and Freund developed early boosting algorithms [5]. Those first methods had a practical problem: they needed to know in advance how good the weak learners were.

That is where "adaptive" comes in. AdaBoost adjusts automatically to the weak learners' actual error rates as it runs, so no such prior knowledge is needed. It does this in two ways:

- **It reweights the training examples each round.** Examples the previous classifiers got wrong are given more weight, so the next weak learner focuses on the hard cases.
- **It weights each weak learner's vote by its accuracy.** More accurate classifiers count for more in the final combined prediction.

So the name says what the algorithm does: it boosts weak learners into a strong one, and it adapts to how well each of them performs as it goes.

## References


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